

# End-to-End Algorithmic Trading Pipeline

## From Raw Price Data to Risk-Adjusted Alpha

# Introduction & Objective

**Goal:** Develop an end-to-end algorithmic trading pipeline for  $N$  anonymized stocks.

**Core Focus:** Transitioning from raw price data to a system that maximizes risk-adjusted returns.

## Stage Checklist:

- Data Engineering
- Strategy Formulation
- Simulation

## What to Explain:

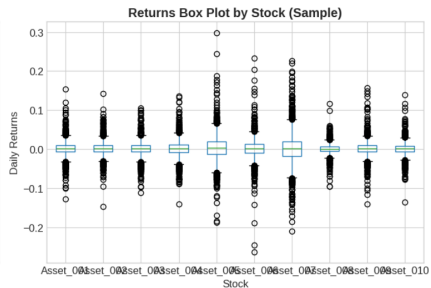
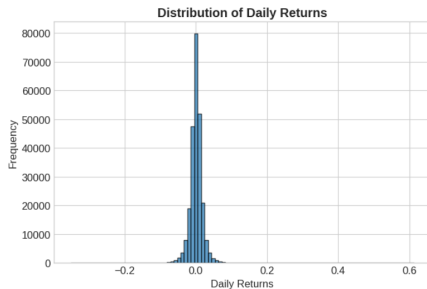
**Dataset:** Daily OHLCV data from Jan 2016 to Jan 2026, totaling 2,511 trading days.

**Cleaning Logic:**

- Fixed OHLC violations (8 High and 11 Low violations)
- Handled missing values using forward/backward fills

**Outlier Management:** Applied Winsorization to clip extreme returns at the 1st and 99th percentiles.

# Distribution of Daily Returns & Box Plot



# Distribution of Daily Returns & Box Plot

**Explanation:** These charts show the raw “noise” in the stock returns. The box plot identifies thousands of individual data points as statistical outliers.

**Insight:** Financial data is fat-tailed. If we don't clip these outliers, the model might mistake a one-day random spike for a tradeable trend.

# Winsorization Impact

Handling Outliers via Winsorization:

|       | Original      | After Winsorization |
|-------|---------------|---------------------|
| count | 251000.000000 | 251000.000000       |
| mean  | 0.000799      | 0.000774            |
| std   | 0.018895      | 0.016725            |
| min   | -0.351166     | -0.092930           |
| 25%   | -0.007424     | -0.007424           |
| 50%   | 0.000816      | 0.000816            |
| 75%   | 0.009144      | 0.009144            |
| max   | 0.612184      | 0.110344            |

# Winsorization Impact

**Insight:** While the Mean stayed almost the same, the Standard Deviation dropped from 0.0189 to 0.0167. This means we made the data cleaner and more stable for the machine learning models without losing the core signal.

Financial data is notoriously messy. We used Z-score and IQR methods to detect thousands of outliers. By using Winsorization, we ensure our models don't overfit but instead learn from the consistent distribution of the market.

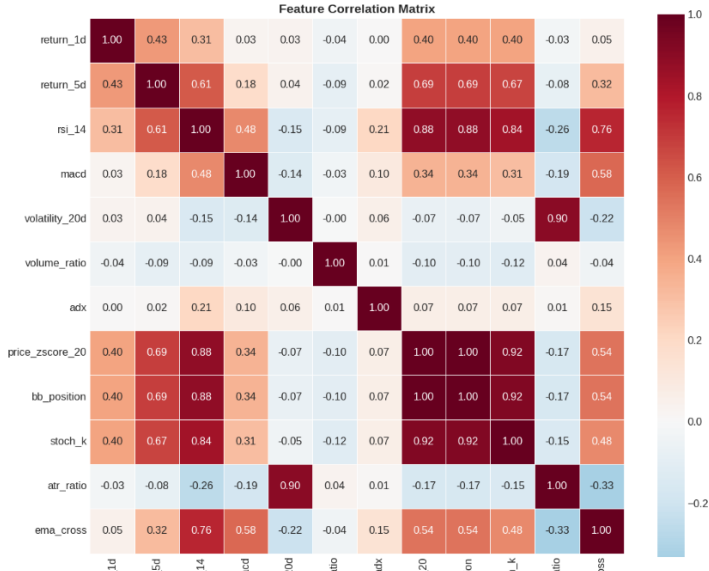
**Indicator Universe:** 57 total features created across 6 categories.

## Key Categories:

- Returns (1d–20d)
- Momentum (RSI, MACD)
- Volatility (ATR, BB Width)
- Volume (OBV)
- Statistical factors (Skew, Kurtosis)



# Feature Correlation Matrix



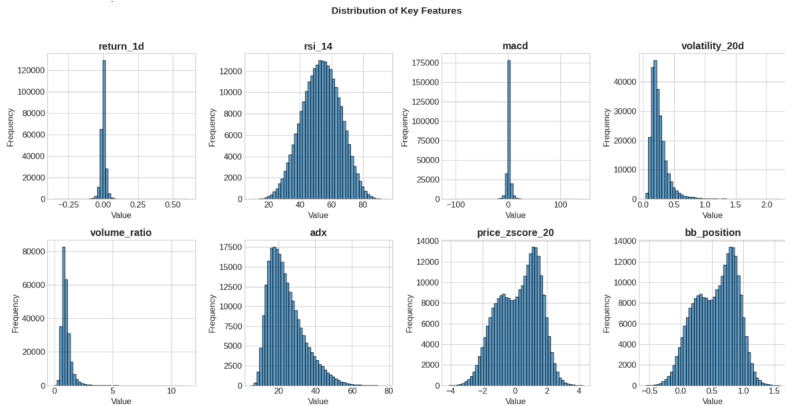
# Feature Correlation Matrix

**Explanation:** A heatmap showing how different technical indicators relate to one another.

**Insight:** Notice the dark red squares — for example, the high correlation between RSI 14 and Price Z-Score.

**Deep Insight:** We want features that are uncorrelated with each other so the model gets fresh information from every input rather than redundant noise.

# Distribution of Key Features



# Distribution of Key Features

**Explanation:** Histograms of indicators like RSI, MACD, and ADX.

**Insight:** Indicators like RSI are normally distributed, which helps the models (especially Neural Networks) converge faster during training.

Simple raw prices are rarely enough. We looked for features with predictive power, such as `overnight_gap` and `volatility_ratio`. We used a Correlation Matrix to ensure our features aren't redundant; for instance, notice how RSI and Stochastic indicators correlate highly, so we balanced our feature set to avoid multi-collinearity.

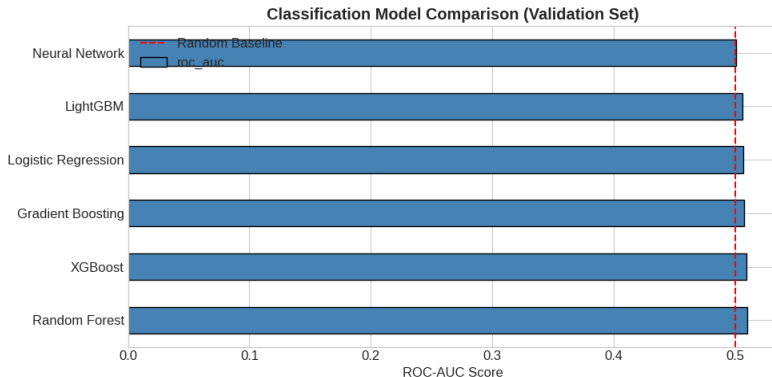
**Predictive Engine:** An ensemble of models including Random Forest, XGBoost, and LightGBM.

**Target:** Predict “Up Probability” and forward returns.

**Ensemble Weights:** Assigned based on ROC-AUC scores (RF and XGBoost at  $\sim 0.168$ ).

# Classification Model Comparison

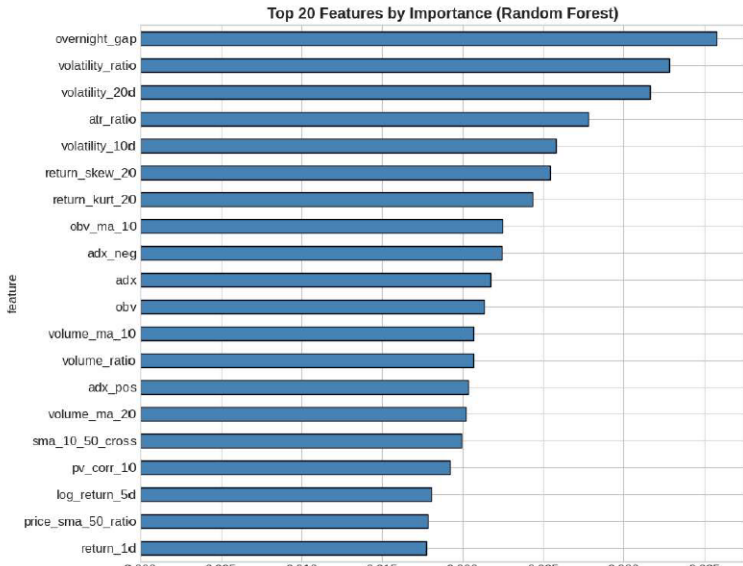
|                            | accuracy | f1       | roc_auc  |
|----------------------------|----------|----------|----------|
| <b>Random Forest</b>       | 0.504085 | 0.504284 | 0.510146 |
| <b>XGBoost</b>             | 0.522825 | 0.474985 | 0.509255 |
| <b>Gradient Boosting</b>   | 0.524736 | 0.458700 | 0.507277 |
| <b>Logistic Regression</b> | 0.505915 | 0.506205 | 0.506449 |
| <b>LightGBM</b>            | 0.524533 | 0.457836 | 0.506221 |
| <b>Neural Network</b>      | 0.510752 | 0.496547 | 0.500930 |



# Classification Model Comparison

**Insight:** Even though the accuracy is just above 50% ( $\sim 0.52$ ), in quantitative finance, even a 1–2% edge over random is enough to be highly profitable when compounded.

# Top 20 Features by Importance





# Top 20 Features by Importance

**Insight:** Overnight Gap and Volatility Ratio are the most important. This tells us that market moves occurring between closing and opening bells are key drivers of alpha in this specific universe.

Markets evolve, and a model that works in Year 1 might fail in Year 3. By using an ensemble, we diversify our model risk. Our importance analysis shows that volatility and volume patterns are key alpha drivers for this universe.

# Backtesting & Performance Analysis

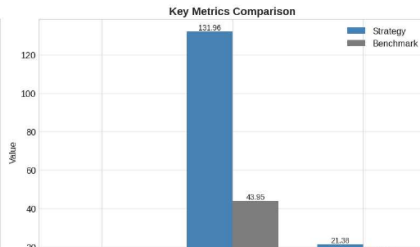
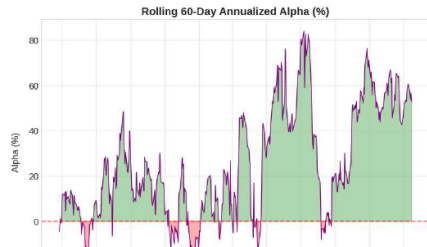
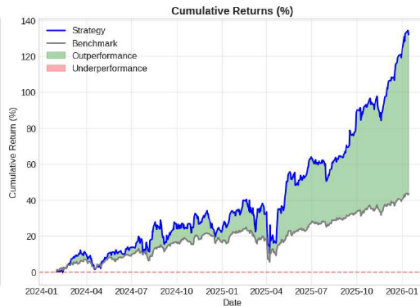
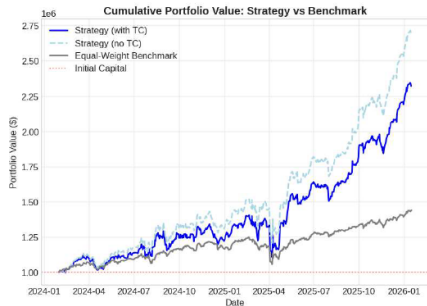
## Parameters:

- 2-year out-of-sample test (Jan 2024 – Jan 2026)
- \$1M capital
- 10 bps transaction costs

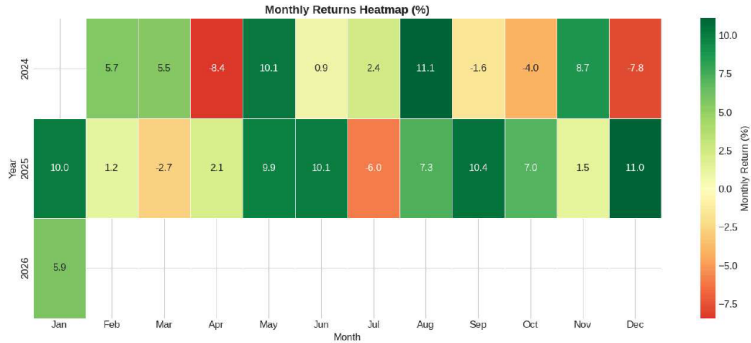
## Results:

- Strategy Return: 131.96%
- Equal-Weight Benchmark: 43.95%
- Sharpe Ratio: 1.98 (Strategy) vs 1.52 (Benchmark)

# Cumulative Portfolio Value



# Monthly Returns Heatmap



# Monthly Returns Heatmap

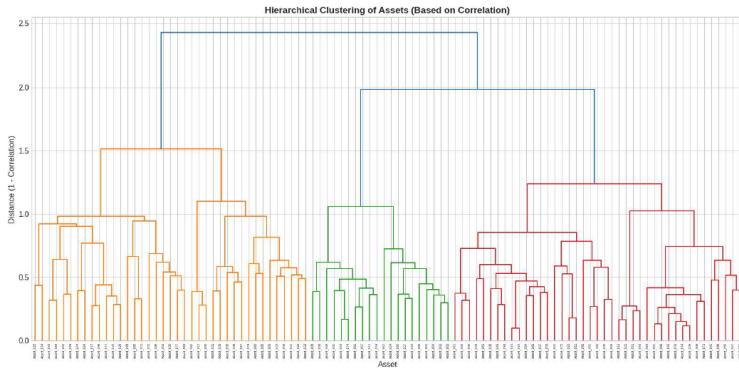
**Insight:** We only had a few worst months (e.g., April 2024 at -8.4%), but they were heavily outweighed by best months exceeding +10%. This consistency is what drives the high Sharpe ratio.

The strategy survived transaction costs totaling over \$209k. While transaction costs reduced total return by roughly 37%, the final Sharpe ratio of 1.98 indicates very high quality, risk-adjusted performance. We rebalanced every 5 days, reducing turnover by nearly 80%.

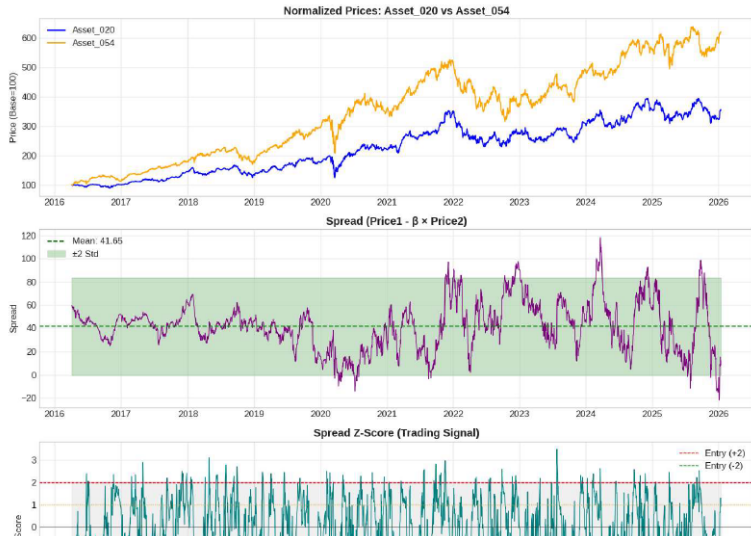
**Relative Value:** Identified cointegrated pairs like Asset\_020 and Asset\_054.

**Verification:** ADF test confirmed the spread is mean-reverting (p-value: 0.0000).

# Hierarchical Clustering Dendrogram



# Spread Z-Score Trading Signal





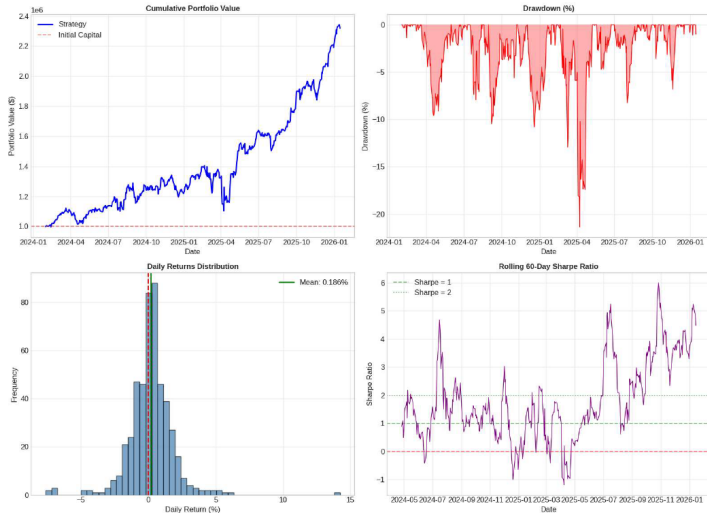
# Spread Z-Score Trading Signal

**Insight:** When the line hits  $+2$ , the pair is stretched. We sell the overpriced one and buy the underpriced one, betting on mean reversion.

Beyond broad alpha, we found specific relative-value opportunities. By trading the spread — selling the winner and buying the loser — we add a market-neutral signal to the main portfolio.

# Final Insights & Verdict

Strategy Performance (With 10 bps Transaction Costs)



# Final Insights & Verdict (Explanation)

**Alpha:** Excess return of 88.01% over the benchmark.

**Stability:** Sharpe ratio  $> 1$  indicates strong risk-adjusted returns.

**Insight:** While returns were  $3\times$  higher, Maximum Drawdown (-21.38%) was kept within a reasonable range compared to the benchmark (-15.56%).

This project shows that a disciplined approach to data cleaning and ensemble modeling can navigate noisy markets.